

An item response theory analysis of the Carver and White (1994) BIS/BAS Scales

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Abstract

The Carver and White BIS/BAS Scales comprise a scale for measuring behavioural inhibition sensitivity and three scales for measuring the behavioural activation sensitivity factors of reward responsiveness, drive and fun seeking. Item response theory (IRT), using Samejima's (1969) graded response model, was used to examine the psychometric properties of these four scales. A total of 520 adult participants completed the BIS/BAS Scales. Although all items in all four scales were reasonably effective in discriminating their respective traits, they provided good measures of their respective latent traits from only moderately low to moderately high trait levels. Also, the category response curves for the first three of the four response options for the items in the reward responsiveness scale showed considerable overlap. These findings suggest some limitations in the psychometric qualities of the BIS/BAS Scale.

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1. Introduction

Gray's (1987) Reinforcement Sensitivity Theory (RST) suggests two independent biological systems that govern behaviour, namely the behavioural inhibition system (BIS) and the behavioural

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approach system (BAS). The BIS/BAS Scales (Carver & White, 1994) have been used widely to tap the personality traits underlying the BAS and BIS. This study used item response theory (IRT) to examine the psychometric properties of the BIS/BAS Scales.

The BIS/BAS Scales have a single scale for the BIS and three BAS Scales. The BIS Scale (seven items) measures BIS sensitivity. The BAS Scales measure reward responsiveness (five items), drive (four items), and fun seeking (four items). Reward responsiveness comprises items reflecting the degree to which rewards lead to positive emotions, while drive comprises items reflecting a person's tendency to actively pursue appetitive goals. Fun seeking reflects items on the tendency to seek out and impulsively engage in potentially rewarding activities. Each item of the BIS/BAS Scales is rated on a four point Likert-type scale, where 1 indicates "strongly disagree" and 4 indicates "strongly agree", with higher scores indicating higher sensitivities.

Past studies based on classical test theory (CTT; Lord & Novick, 1968) have generally supported the factor structure, reliability and validity of the BIS/BAS Scales (Carver & White, 1994; Caseras, Avila, & Torrubia, 2003; Gomez & Gomez, 2005; Heubeck, Wilkinson, & Colongon, 1998; Jorm et al., 1999; Ross, Millis, Bonebright, & Bailley, 2002; Zelenski & Larsen, 1999). It has been argued that item response theory (IRT) has many advantages over CTT for evaluating the psychometric properties of measures (Embretson & Reise, 2000; Steinberg & Thissen, 1996). Unlike CTT, IRT approaches provide a trait score for each item, reliability of each item at different levels of the underlying trait, and the psychometric properties obtained are sample or group invariant.

The BIS/BAS Scales have a polytomous response format (i.e., four-point rating scale), with the response options graded. An IRT model appropriate for this item response format is Samejima's (1969) graded response model (GRM). For this model, logistic curves, called category response curves (CRCs) are generated for each response option with an item, showing the probability of a positive response to the option as a function of the underlying trait, with the slopes for all options in each item constrained equal. Thus the GRM will estimate only one discrimination parameter (α). The α value indicates the ability of an item to discriminate people of different levels of the underlying trait. According to Baker (2001), α values 0.01–0.24 are very low, 0.25–0.64 are low, 0.65–1.34 are moderate, 1.35–1.69 are high, and more than 1.7 are very high. In addition, for each item it is also possible to derive threshold parameters indicating the trait level (q) where there is a 0.5 probability of endorsing the relevant response option or higher response options. The number of threshold parameters per item will be the number of response options minus 1. The GRM also provides values for item information function and test information function. The item information function indicates the reliability of an item to measure the trait at different levels of the trait, while the test information function provides the reliability of the test (i.e., all items together) to measure the trait at different levels of the trait. The GRM also provides the standard error of measurement (SEM) of the test information functions, which is the inverse of the reliability of the test, and thus can provide information on the precision of the test (Embretson & Reise, 2000).

At present no study that we are aware of has used IRT to examine the psychometric properties of the BIS/BAS Scales. Thus the aim of the study was to use the GRM to determine their psychometric properties. Given existing data and Carver and White's (1994) model of the BIS/BAS Scales, we used IRT to examine the psychometric properties of the items within each of the four scales (i.e. BIS, BAS reward responsiveness, BAS drive and BAS fun seeking) separately.

2. Method

2.1. Participants

The participants comprised 520 individuals. Participants' age ranged from 18 to 42 years, with a mean age of 21.46 (S.D. = 6.05). There were 204 male and 316 female participants. The mean age for the male participants was 22.62 years (S.D. = 4.79), and was 21.17 years (S.D. = 7.32) for female participants. Participants were recruited from the University of Ballarat, Australia, through advertisements and leaflets placed on notice boards around the university. Only individuals who provided informed written consent participated in the study.

2.2. Measures

The BIS/BAS Scales have already been described earlier. In brief, each item of the BIS/BAS Scales is rated on a four point Likert-type scale, where 1 indicates “strongly disagree” and 4 indicates “strongly agree”, with higher scores indicating higher sensitivities.

2.3. Procedure

Participants were tested in groups of 5–20 in a laboratory. After first providing their informed consent to participate, they were told that the research was aimed at examining a particular personality measure. Following this, they were asked to complete the personality questionnaire. When all participants in the session completed the questionnaire, they returned the questionnaire to the supervising researcher, and were debriefed on the nature of the study.

2.4. Statistical procedures

This study used Multilog 7.0.3 (Thissen, 1991) to perform Samejima (1969) GRM IRT analyses. As with most IRT models, the GRM assumes that the items in a questionnaire form a single dimension (unidimensionality). Implied in this assumption is that responses to items are independent of one another or local independence. This means that other than the underlying trait there is no association between the items. Also, as GRM is model-based, it is necessary to test if there is model-data fit. Thus prior to the IRT analyses, the unidimensionality, local independence and model-data fit of the items were evaluated for each of the four subscales.

3. Results

3.1. Unidimensionality and local independence

The results for the tests for unidimensionality of the four BIS/BAS Scales are shown in Table 1. For all four scales, the mean corrected item-to-total correlations were high. The results from the principal components exploratory factor analysis (EFA), involving oblique rotation, showed only

one factor for each scale with an eigenvalue of more than one. For these scales, the ratios of the eigenvalues of the first and second factors from the EFA showed noticeably high values for the first factor extracted compared to the second factor extracted. The results of the confirmatory factor analysis (CFA), involving robust analysis of the covariance matrix with maximum likelihood estimation procedure, are also shown in Table 1. These findings indicated excellent fit for the one-factor models. Taken together, these findings support the unidimensionality of each scale, and also the local independence of the items in each scale (Chen & Thissen, 1997). Table 1 also shows the Cronbach's alpha values for all scales. The values range from 0.69 to 0.82, indicating that all scales had high internal consistency.

3.2. Model-data fit

For each item, MULTILOG provides the observed proportion and the expected proportion of the responses in each category. The expected proportion is predicted from the item parameters and the estimated latent trait distribution. Thus the difference in the proportions or residual can be used to examine model-data fit, with bigger differences indicating lesser model-data fit. The residuals for all items of all scales were mostly 0.00 and only one value was greater than 0.01 (the item 'I go out of my way to get things I want' had a value of 0.02). This indicates excellent model-data fit.

3.3. IRT analyses of the scales of the BIS/BAS scales

Table 2 shows the discrimination and the threshold parameter values for all items of the four scales of the BIS/BAS Scales. Table 3 shows the item and test information values, and the reliabil-

Table 1
Homogeneity of the four scales of the BIS/BAS Scales

	BIS	BAS-RR	BAS-D	BAS-FS
Mean item-to-total correlations	0.69	0.67	0.81	0.77
Exploratory factor analysis				
Eigenvalues of factor 1	3.34	2.23	2.61	2.36
Eigenvalues of factor 2	0.87	0.79	0.56	0.66
Ratio of eigenvalues of factor 1 to factor 2	3.84:1	2.8:1	4.67:1	3.58:1
Confirmatory factor analysis—fit indices				
Satorra–Bentler chi square	59.16*	4.90	5.62	9.94*
Root mean square error of approximation	0.08	0.00	0.06	0.09
Comparative fit index	0.94	1.00	0.99	0.98
Standardized root mean square residual	0.03	0.01	0.01	0.02
Standardized loading—range	0.53–0.73	0.49–0.67	0.64–0.88	0.58–0.76
Standardized loading—median value	0.61	0.56	0.71	0.68
Internal consistency (Cronbach's alpha)	0.82	0.69	0.82	0.76

Note: BIS = BIS Scale; BAS-RR = BIS reward responsiveness scale; BAS-D = BAS drive scale; BAS-FS BAS fun seeking scale.

* $p < 0.05$.

Table 2
IRT parameter estimates for the Carver and White BIS/BAS Scale

	Item parameter estimates			
	α	β_1	β_2	β_3
<i>BIS Scale</i>				
If I think something unpleasant is going to happen I usually get pretty worked up	1.68 (0.16)	−2.62 (0.25)	−0.57 (0.09)	1.37 (0.14)
I worry about making mistakes	1.64 (0.16)	−1.95 (0.20)	−0.66 (0.09)	1.36 (0.14)
Criticism or scolding hurts me quite a bit	1.87 (0.15)	−2.18 (0.20)	−0.55 (0.09)	1.21 (0.11)
I feel pretty worried or upset when I think or know somebody is angry at me	2.32 (0.20)	−2.30 (0.22)	−1.00 (0.08)	0.85 (0.08)
Even if something bad is about to happen to me, I rarely experience fear or nervousness	1.67 (0.17)	−2.89 (0.34)	−1.42 (0.14)	0.81 (0.11)
I feel worried when I think I have done poorly at something	1.63 (0.13)	−2.84 (0.26)	−1.04 (0.12)	1.52 (0.15)
I have very few fears compared to my friends	1.29 (0.14)	−2.37 (0.28)	−0.47 (0.11)	2.07 (0.22)
<i>Reward Responsiveness Scale</i>				
When I get something I want I feel excited and energised	2.16 (0.21)	−5.61 (0.00)	−2.25 (0.40)	0.26 (0.07)
When I'm doing well at something, I love to keep at it	1.55 (0.23)	−6.71 (0.00)	−2.71 (0.48)	0.00 (0.19)
When good things happen to me, it affects me strongly	1.63 (0.17)	−4.20 (0.86)	−2.45 (0.24)	0.28 (0.09)
It would excite me to win a contest	1.26 (0.17)	−7.40 (0.00)	−3.00 (0.68)	−0.15 (0.11)
When I see an opportunity for something I like, I get excited right away	1.32 (0.16)	−4.31 (0.94)	−2.33 (0.27)	0.64 (0.12)
<i>Drive Scale</i>				
When I want something, I usually go all out to get it	5.35 (0.52)	−2.02 (0.11)	−0.28 (0.04)	0.97 (0.05)
I go out of my way to get things I want	2.82 (0.23)	−2.13 (0.16)	−0.25 (0.06)	1.37 (0.10)
If I see a chance to get something I want, I move on it right away	1.76 (0.18)	−3.49 (0.50)	−0.080 (0.10)	1.14 (0.12)
When I go after something I use a 'no holds barred' approach	1.71 (0.18)	−2.37 (0.23)	−0.12 (0.08)	1.76 (0.15)
<i>Fun Seeking Scale</i>				
I will often do things for no other reason than that they might be fun	1.60 (0.16)	−2.74 (0.31)	−1.17 (0.12)	1.00 (0.12)
I crave excitement and new sensations	2.68 (0.23)	−2.62 (0.26)	−0.87 (0.07)	0.67 (0.07)
I'm always willing to try something new if I think it will be fun	2.51 (0.22)	−2.75 (0.29)	−1.41 (0.10)	0.52 (0.08)
I often act on the spur of the moment	1.62 (0.15)	−2.25 (0.21)	−0.33 (0.09)	1.26 (0.13)

Note: α = discrimination parameter; β_1 , β_2 and β_3 = threshold parameters.

Table 3

Item and test information functions for the four BIS/BAS Scales at various levels of the trait

Items	Estimated trait						
	–3.0	–2.0	–1.0	0	1.0	2.0	3.0
<i>BIS Scale</i>							
Worked up	0.64	0.66	0.70	0.69	0.71	0.54	0.16
Mistakes	0.35	0.72	0.77	0.64	0.68	0.52	0.16
Criticism	0.51	0.91	0.88	0.84	0.90	0.53	0.12
Angry at me	0.74	1.38	1.42	0.91	1.32	0.33	0.04
Fear/nervousness	0.72	0.76	0.68	0.59	0.69	0.30	0.07
Poorly	0.62	0.70	0.71	0.48	0.60	0.57	0.20
Few fears	0.36	0.46	0.46	0.43	0.39	0.43	0.30
TIF	4.96	6.60	6.62	5.57	6.28	4.42	2.04
SEM	0.45	0.39	0.39	0.42	0.40	0.49	0.70
R	0.80	0.85	0.85	0.82	0.84	0.76	0.51
<i>Reward Responsiveness Scale</i>							
Excited/energised	0.66	1.10	0.51	1.10	0.65	0.10	0.01
Keep at it	0.58	0.50	0.45	0.61	0.35	0.10	0.02
Affects	0.68	0.61	0.42	0.65	0.48	0.14	0.03
Contest	0.41	0.35	0.36	0.40	0.24	0.09	0.03
Excited right away	0.47	0.44	0.33	0.40	0.41	0.21	0.07
TIF	3.79	4.00	3.07	4.15	3.14	1.65	1.17
SEM	0.51	0.50	0.57	0.49	0.57	0.78	0.93
R	0.74	0.75	0.68	0.76	0.68	0.39	0.14
<i>Drive Scale</i>							
Go all out	0.15	7.14	0.70	4.38	7.11	0.11	0.00
Get things	0.58	1.94	1.00	1.84	1.65	0.99	0.08
Move on it	0.68	0.46	0.78	0.69	0.80	0.46	0.11
No holds barred	0.56	0.71	0.59	0.77	0.69	0.71	0.28
TIF	2.97	11.94	4.06	8.68	11.24	3.27	1.47
SEM	0.58	0.30	0.50	0.34	0.30	0.55	0.83
R	0.66	0.91	0.75	0.88	0.91	0.70	0.31
<i>Fun Seeking Scale</i>							
Might be fun	0.63	0.69	0.68	0.56	0.66	0.36	0.10
New sensations	1.41	1.18	1.79	1.30	1.49	0.19	0.01
Something new	1.44	1.41	1.29	1.17	1.11	0.15	0.01
Spur	0.47	0.69	0.64	0.72	0.71	0.47	0.14
TIF	4.95	4.96	5.41	4.75	4.98	2.17	1.26
SEM	0.45	0.45	0.43	0.46	0.45	0.68	0.89
R	0.80	0.80	0.82	0.79	0.80	0.54	0.21

Note: TIF = test information function; SEM = standard error of measurement; R = reliability.

ity of the four scales at different levels of their latent traits. For illustrative purposes, column 1 of Fig. 1 shows the category response curves (CRCs) for an item in each scale, while column 2 of this figure shows the corresponding item information curves. Column 3 of this figure shows the test information curves for the four scales of the BIS/BAS Scales.

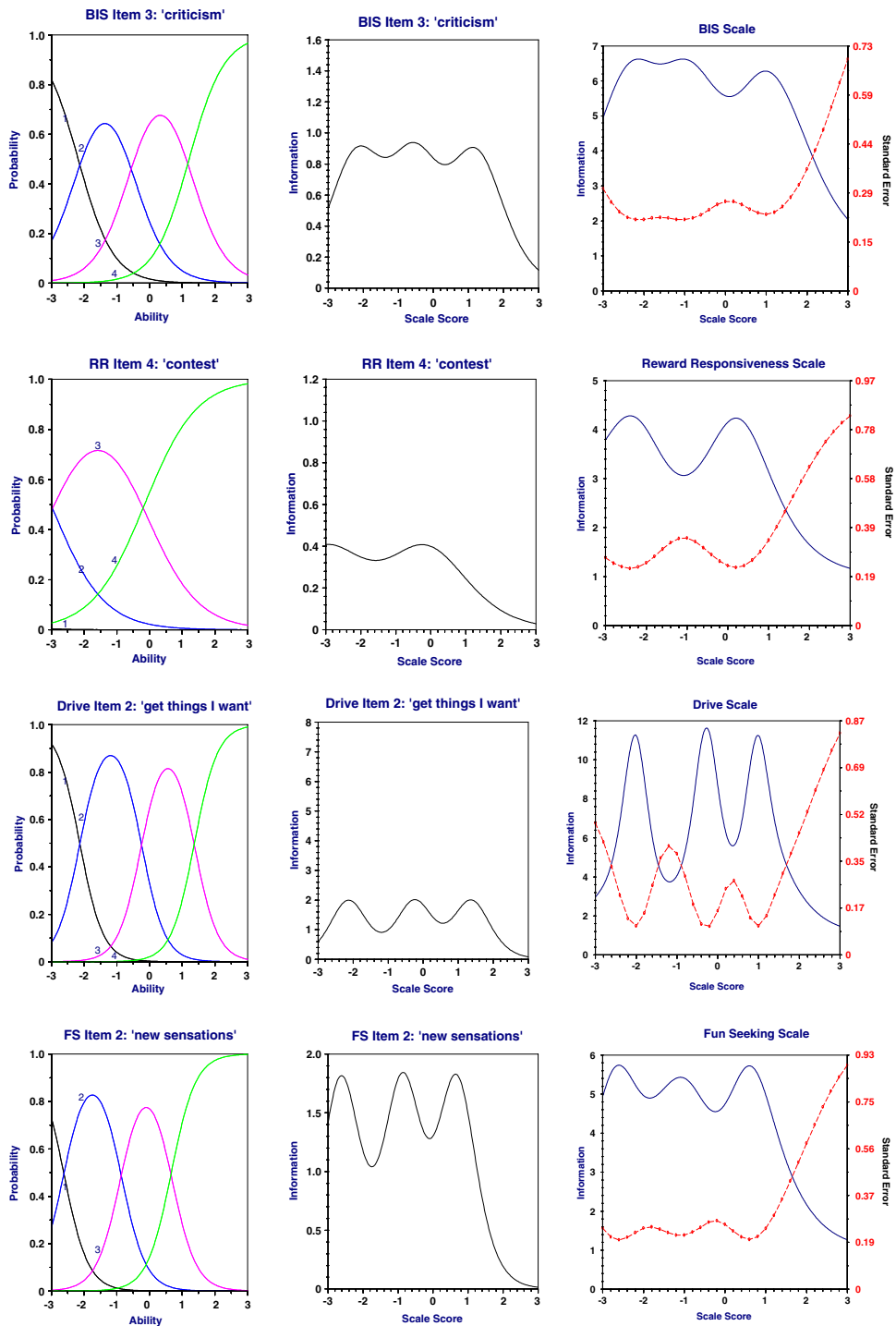


Fig. 1. Items illustrating category response curves (CRCs) and item information function and the test information function and standard error (dotted line) of the four BIS/BAS Scales.

3.4. IRT analyses of the BIS Scale

Table 2 shows that the α value for “few fears” was moderate, while the α values for the other items were either high or very high. The β_1 values for all items were around or less than 2 S.D. below the mean trait level, while the β_2 values were around 0.4 S.D. and 1 S.D. below the mean trait level. Except for “poorly” and “few fears”, the β_3 values were around 1 S.D. above the mean. For “poorly” and “few fears”, the β_3 values were around 1.5 S.D. and 2 S.D. above the mean trait level, respectively.

Inspection of the CRCs of the BIS items showed that there was fairly good separation in the response options. In general, for all seven BIS items, the first and second response options covered mostly the area below the mean trait level, while the third and fourth response options covered mostly the area above the mean trait level (see row 1 column 1 of Fig. 1 for an illustration).

From Table 3 and row 1 column 3 of Fig. 1, it will be noticed that for the BIS Scale as a whole, the S.E., test information function and reliability levels were good for trait levels from 3 S.D. below the mean trait level to 2 S.D. above the mean trait level. Also, all items showed reasonable item information function (and therefore high reliability) values between these trait levels, with this being especially good for the item “angry at me”.

3.5. IRT analyses of the BAS reward responsiveness Scale

As can be seen in Table 2, the α parameter values showed some variability, with this value being very high for “excited/energised”, high for “keep at it” and “affects”, and moderate for “contest” and “excited right away”. This table shows that for β_1 , all items had very low values, ranging from 4.20 S.D. to 7.40 S.D. below the mean trait level. Their β_2 values were also low, ranging from 2.25 S.D. to 3.00 S.D. below the mean trait level. The β_3 values were around the mean trait level.

Investigation of the CRCs for each item showed that, except for the item “excited/energised”, there was considerable overlap between response options one, two and three, for the other four items. For “excited/energised” there was mainly overlap between response options one and two. The CRCs also showed that for all items the first, second and third response options covered the area mainly below the mean trait level, with the fourth response option covering the area slightly below and above the mean trait level (see row 2 column 1 of Fig. 1 for an illustration).

From Table 3 and row 2 column 3 of Fig. 1, it will be noticed that for the BAS reward responsiveness scale as a whole, the S.E., test information function and reliability levels were reasonably good for the trait levels around the mean and around 3 S.D. to 1.5 S.D. below the mean. They were also generally acceptable for trait levels around 1 S.D. below and above the mean trait level. There was, however, a degree of variability in the item information function (and therefore reliability). Although all five items tended to show increased information values at a point around the mean to 1.0 S.D. above the mean trait level, the items on “contest” and “excited right away” showed relatively low information values, the items on “keep at it” and “affects” showed moderate information values, and the item on “excited/energised” showed relatively high information values, particularly between 2.0 S.D. below the mean and the mean trait level.

3.6. IRT analyses of the BAS drive Scale

Table 2 shows that all the α parameter values were very high. For β_1 , the item “move on it” had a value around 3.5 S.D. below the mean trait level, while the values for the other items were closer to 2 S.D. below the mean trait level. With the exception of the item “move on it”, for all the other items, the β_2 values were around the mean trait level. The β_2 value for “move on it” was closer to 1 S.D. below the mean. The β_3 values for all items were close to or above 1 S.D. above the mean, with it being closer to 2 S.D. for the item “no holds barred”.

The CRCs for each item showed that, for the item “If I see a chance to get something I want, I move on it right away” there was considerable overlap for the different response options. For the remaining three items, there was generally good separation of the different response options. The CRCs, however, showed that for all four items, the first and second response options covered mainly the area below the mean trait level, while the third and fourth response options covered a relatively large area for trait levels above the mean (see row 3 column 1 of Fig. 1 for an illustration).

As shown in Table 3 and row 3 column 3 of Fig. 1, for the drive scale as a whole, the S.E., the test information function and reliability levels were generally good for trait levels between 2 S.D. below and above the mean trait level. Also, all items showed reasonable information values between these trait levels. Also, the item information function (and therefore reliability) values for the items “go all out” and “gets things” were higher, relative to other items.

3.7. IRT analyses of the BAS fun seeking scale

As shown in Table 2, the α parameter values for the items in this scale were either high or very high. As will be noticed in this table, the β_1 values for all items were around 2.5 S.D. below the mean trait level. The β_2 value for “spur” was around the mean trait level, while this value for the other items were closer to 1 S.D. below the mean trait level. The β_3 values for all items were between 0.5 S.D. and 1 S.D. above the mean trait level.

The CRCs for each item show that there was generally good separation of the different response options. The CRCs showed that for all four items, the first and second response options covered the area from low to slightly above the mean trait level, while the third and fourth response options covered mainly the area for trait levels above the mean (see row 4 column 1 of Fig. 1 for an illustration).

For the fun seeking scale as a whole, the S.E., test information function and reliability levels were good for trait levels between 1 S.D. above the mean and 3 S.D. below the mean (see Table 3 and row 4 column 3 of Fig. 1). Also, all items showed reasonable item information values at these levels. Despite this, the items “might be fun” and “spur” showed relatively moderate information values from 3.0 S.D. below the mean trait to 2.0 S.D. above the mean trait level, whilst the other two items (“new sensations” and “something new”) showed relatively higher information values for these trait levels.

4. Discussion

This study used the Graded Response Model (GRM) to examine the item response theory (IRT) psychometric properties of the items in the BIS, BAS reward responsiveness, BAS drive, and BAS fun seeking scales of the BIS/BAS Scales. In relation to the discrimination parameters,

with the exception of the BIS item “few fear” and the BAS items “contest” and “excited right away”, the discrimination parameter values of all other items in all four scales were either high or very high. The discrimination parameter values for “few fear”, “contest” and “excited right away” were all in the moderate range. The results also showed that generally all the BIS items provide good representation (as indicated by the threshold parameters) and reliability (as indicated by the item information functions) of the BIS trait starting around 2 S.D. below the mean to 1 S.D. above the mean, with “few fears”, representing this trait with reasonable reliability up to around 2 S.D. above the mean. The results showed that the BAS reward responsiveness items provide good representation of this trait and has reasonable reliability between extremely low levels to the mean levels of this trait. In relation to the BAS drive trait, the results indicated that the items in the BAS Drive scale provide good representation and good reliability from this trait at least 2 S.D. below the mean to around 1 S.D. above the mean, with “no holds barred”, representing this trait with reasonable reliability up to around 2 S.D. above the mean. All the BAS fun seeking items were able to represent well and provide good reliable measures for the BAS fun seeking trait from around 2 S.D. below the mean to around 0.5 to 1 S.D. above the mean.

Overall, these findings indicate that, with the exception of the BIS item “few fears”, all the other items of the BIS, BAS drive and BAS fun seeking scales are able to discriminate their respective traits reasonably well. Also all items in the three scales provide reasonable representation and reliable measures of their respective traits from at least 2 S.D. below to about 1 S.D. above the respective mean trait levels. In addition to these findings, the threshold parameter values for all items in all three scales generally showed noticeable increases in the level of their respective traits for each response option, and there was generally clear separation between response options.

Taken together, these findings imply that the items in the BIS, BAS drive and BAS fun seeking scales are good measures of their respective traits at only low to slightly above the appropriate mean trait levels. This means that these items are poor for measuring the appropriate traits at moderately high to high levels. This implies that most of the items in the BIS, BAS drive and BAS fun seeking scales are limited. Although BIS “few fears” showed reliable and good representation of the BIS trait at even higher levels, it was only moderate for discriminating this trait. Thus there is a limitation in this item as well. It would appear that of all the items across the BIS, BAS drive and BAS fun seeking scales, only the BAS drive item “no holds barred” has reasonable item response characteristics, in that it has a high discrimination parameter, its threshold parameters range from about 2 S.D. below to 2 S.D. above the mean, and there is clear separation in its CRCs.

The results for the BAS reward responsiveness scale showed that at least two of its five items (i.e., “contest” and “excited/energised”) had only moderate discrimination ability and low item information values. Also, all items were only effective in representing the BAS reward responsiveness trait at very low to just around the mean trait levels. Furthermore, the CRCs showed for all items there was considerable overlap between response options. Thus relative to the BIS, BAS drive and BAS fun seeking scales, the BAS reward responsiveness scales showed more serious limitations.

Overall then, it appears that from an IRT perspective, the four scales of the BIS/BAS Scales have limitations. The findings here do not provide as strong a support for the psychometric properties for the BIS/BAS Scales, as provided from a CTT perspective. The findings in this study indicated a number of improvements needed for the BIS/BAS Scales. First, there is need to improve the discrimination ability of the BIS item “few fears” and the BAS items “contest” and “excited right away”. Second, there is a need to revise or modify the items so that they represent and reliably mea-

sure the appropriate traits at moderately high and high levels also. Third, for the items in the BAS reward responsiveness scale, there is a need to revise the items so that the response options are more clearly differentiated when responded to. Alternatively, it may be necessary to reduce the number of response options for the items in this scale, such as requiring “yes” or “no” responses only. Clearly, the BIS/BAS Scale is in need of revision. IRT analysis can play a valuable role in this process.

In concluding, this study has shown how the use of IRT procedures can provide valuable additional psychometric information on the BIS/BAS Scales, and also how this procedure can provide useful practical guidelines for improving the design of this questionnaire. The IRT methodology is appropriate for questionnaires in general. We hope that our study will encourage other researchers to use IRT to evaluate the psychometric properties of other commonly used self-rating personality questionnaires, in particular the large number of questionnaires that are currently used in RST studies. Such studies would be valuable as they will facilitate the development of bench-mark questionnaire measures of the BIS and BAS. This is certainly needed, given this is not available at present, and that theoretical developments in this area have made great gains in recent years.

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